

MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS
NANYANG TECHNOLOGICAL UNIVERSITY

Tutorial 4

HINTS

.....
 We will discuss several questions from Chapter 4 in Ross' book.

Tutorial Questions

- (1) Let X be a binomial random variable with parameters n and p . Show that

$$\mathbb{E}\left(\frac{1}{X+1}\right) = \frac{1 - (1-p)^{n+1}}{(n+1)p}.$$

Proof:

$$\begin{aligned} \mathbb{E}\left(\frac{1}{X+1}\right) &= \sum_{k=0}^n \frac{1}{k+1} \binom{n}{k} p^k (1-p)^{n-k} \\ &= \sum_{k=0}^n \frac{1}{n+1} \binom{n+1}{k+1} p^k (1-p)^{n-k} \\ &= \frac{1}{n+1} \sum_{j=1}^{n+1} \binom{n+1}{j} p^{j-1} (1-p)^{n+1-j} \\ &= \frac{1}{(n+1)p} \sum_{j=1}^{n+1} \binom{n+1}{j} p^j (1-p)^{n+1-j} \\ &= \frac{1}{(n+1)p} \left(\sum_{j=0}^{n+1} \binom{n+1}{j} p^j (1-p)^{n+1-j} - (1-p)^{n+1} \right) \\ &= \frac{1}{(n+1)p} (1 - (1-p)^{n+1}). \end{aligned}$$

A crucial point

$$\begin{aligned} \bullet \quad \frac{1}{k+1} \binom{n}{k} &= \frac{1}{n+1} \binom{n+1}{k+1} \text{ by } \frac{1}{k+1} \binom{n}{k} = \frac{1}{k+1} \cdot \frac{n(n-1)\cdots(n-k+1)}{k!} \\ &= \frac{1}{n+1} \cdot \frac{(n+1)n(n-1)\cdots(n-k+1)}{(k+1)k!} = \frac{1}{n+1} \cdot \frac{(n+1)!}{((n+1)-(k+1))(k+1)!} = \frac{1}{n+1} \binom{n+1}{k+1}. \end{aligned}$$

- (2) Each of 500 soldiers in an army company independently has a certain disease with probability $1/10^3$. This disease will show up in a blood test, and to facilitate matters, blood samples from all 500 soldiers are pooled and tested.

- (a) What is the (approximate) probability that the blood test will be positive (that is, at least one person has the disease)?

Suppose now that the blood test yields a positive result.

- (b) What is the probability, under this circumstance, that more than one person has the disease?

One of the 500 people is Jones, who knows that he has the disease.

- (c) What does Jones think is the probability that more than one person has the disease?

Because the pooled test was positive, the authorities have decided to test each individual separately. The first $i - 1$ of these tests were negative, and the i th one—which was on Jones—was positive.

- (d) Given the preceding scenario, what is the probability, as a function of i , that any of the remaining people have the disease?

Solution: This is a binomial distribution with parameters $n = 500$ and $p = 0.001$, and then, the answer for part (a) is

$$P\{Z > 0\} = 1 - P\{Z = 0\} = 1 - \binom{500}{0} \left(\frac{1}{1000}\right)^0 \left(1 - \frac{1}{1000}\right)^{500}.$$

As $n = 500 > 50$ and $p = 0.001$, $np = 0.5 < 5$, we can use Poisson distribution ($X \sim \text{Poisson}(0.5)$) to approximate this binomial distribution. This is an example of applying Poisson distribution to approximate binomial distributions.

(a)

$$P\{X > 0\} = 1 - P\{X = 0\} = 1 - e^{-0.5} \approx 0.3935.$$

- (b) This is approximately the conditional probability

$$P\{X > 1 | X \geq 1\} = \frac{P\{X > 1\}}{P\{X \geq 1\}} = \frac{1 - e^{-0.5} - 0.5e^{-0.5}}{1 - e^{-0.5}} \approx 0.229$$

- (c) Since Jones knows he has the disease, the positive test result provides no new information to him. The probability in question is just the probability that at least one of the remaining 499 people has the disease, which is (letting $Y \sim \text{Poisson}(0.499)$)

$$P\{Y \geq 1\} = 1 - e^{-0.499} \approx 0.3929.$$

- (d) The desired probability is just the probability that at least 1 among the remaining $500 - i$ people has the disease, which can be approximated by a Poisson random variable with parameter $(500 - i)/1000$ as

$$1 - e^{-0.5 + 0.001i}.$$

- (3) Suppose that 10 balls are put into 5 boxes, with each ball independently being put in

box i with probability p_i , $\sum_{i=1}^5 p_i = 1$.

- (a) Find the expected number of boxes that do not have any balls.
 (b) Find the expected number of boxes that have exactly 1 ball.

Solution:

- (a) Let Z be a random variable for the number of boxes that do not have any balls and $Z_i \in \{0, 1\}$ be the indicator variable whether the i -th box is empty, then $Z_1 + Z_2 + \cdots + Z_5$ is the number of empty boxes. Then $Z = Z_1 + Z_2 + \cdots + Z_5$, and hence

$$\mathbb{E}(Z) = \mathbb{E}(Z_1 + Z_2 + \cdots + Z_5) = \sum_{i=1}^5 \mathbb{E}(Z_i).$$

Note that

$$\begin{aligned} \mathbb{E}(Z_i) &= 1 \cdot P(\text{i-th box is empty}) + 0 \cdot P(\text{i-th box is not empty}) \\ &= P\{\text{i-th box is empty}\} = (1 - p_i)^{10}, \end{aligned}$$

and thus, we have

$$\mathbb{E}(Z) = \sum_{i=1}^5 \mathbb{E}(Z_i) = \sum_{i=1}^5 P\{\text{i-th box is empty}\} = \sum_{i=1}^5 (1 - p_i)^{10}.$$

- (b) Let Y be a random variable for the number of boxes that contains exactly one ball, and $Y_i \in \{0, 1\}$ be the indicator variable whether the i -th box contains exactly 1 ball, then $Y = Y_1 + Y_2 + \cdots + Y_5$ and

$$\mathbb{E}(Y) = \mathbb{E}(Y_1 + Y_2 + \cdots + Y_5) = \sum_{i=1}^5 \mathbb{E}(Y_i).$$

Again, we have $\mathbb{E}(Y_i) = P\{i\text{-th box contains exactly 1 ball}\} = \binom{10}{1} p_i (1 - p_i)^9$, and thus,

$$\mathbb{E}(Y) = \sum_{i=1}^5 10 p_i (1 - p_i)^9.$$

- (4) There are k types of coupons. Independently of the types of previously collected coupons, each new coupon collected is of type i with probability p_i , $\sum_{i=1}^k p_i = 1$. If n coupons are collected, find the expected number of distinct types that appear in this set. (That is, find the expected number of types of coupons that appear at least once in the set of n coupons.)

Sanity checks: What is your answer when $n = 1$? When $n = 2$? When $n \rightarrow \infty$? Why do they make sense?

Solution: Let X_i be the indicator variable whether the type i coupon appears and X be the total number of distinct types of coupons that are collected, then $X = \sum_{i=1}^k X_i$. Hence

$$\mathbb{E}(X) = \mathbb{E}\left(\sum_{i=1}^k X_i\right) = \sum_{i=1}^k \mathbb{E}(X_i) = \sum_{i=1}^k (1 - (1 - p_i)^n).$$

Sanity checks:

- When $n = 1$, $\mathbb{E}(X) = \sum_{i=1}^k p_i = 1$. This is obviously true since I always see some new type in my first coupon, so $X = 1$.
- When $n = 2$, $\mathbb{E}(X) = \sum_{i=1}^k (2p_i - p_i^2) = 2 - \sum_{i=1}^k p_i^2$. The second term $q = \sum_{i=1}^k p_i^2$ is exactly the probability that both coupons belong to the same type. If that happens, I have only 1 type of coupon, otherwise I have 2 types: with probability $1 - q$. The expected number of distinct types can also be calculated as $1 \cdot q + 2 \cdot (1 - q) = 2 - q$, which agrees with our preceding answer.
- When $n \rightarrow \infty$, $(1 - p_i)^n$ has limit 0 and hence $\mathbb{E}(X) \rightarrow \sum_{i=1}^k 1 = k$, and this makes sense that I expect to collect all types of coupons when $n \rightarrow \infty$.

- (5) If X is a geometric random variable, show analytically that

$$P\{X = n + k | X > n\} = P\{X = k\}$$

Using the interpretation of a geometric random variable, give a verbal argument as to why the preceding equation is true.

Solution: Assume that $k \geq 1$, and we calculate as follows:

$$\begin{aligned}
 P\{X = n + k | X > n\} &= \frac{P\{X = n + k \text{ and } X > n\}}{P\{X > n\}} \\
 &= \frac{P\{X = n + k\}}{P\{X > n\}} \\
 &= \frac{(1-p)^{n+k-1}p}{\sum_{i=n+1}^{\infty} (1-p)^{i-1}p} \\
 &= \frac{(1-p)^{n+k-1}p}{\frac{(1-p)^n}{p} \cdot p} \\
 &= (1-p)^{k-1}p \\
 &= P\{X = k\}.
 \end{aligned}$$

An intuitive explanation is that once you know the first n experiments are failed, meaning that you are conditioning on this, you can pretend that the $(n+1)$ -st experiment is now the first one, or, you shift your starting point by n experiments.

- (6) Find the expected value of the sum obtained when n fair dice are rolled.

Solution: Let X be the sum, and we will compute $\mathbb{E}(X)$. We let X_i be the upturned value on die i . Then

$$X = X_1 + X_2 + \cdots + X_n.$$

Because X_i is equally likely to be any of the values from 1 to 6, it follows that

$$\mathbb{E}(X_i) = 1 \left(\frac{1}{6}\right) + 2 \left(\frac{1}{6}\right) + \cdots + 6 \left(\frac{1}{6}\right) = \frac{21}{6} = \frac{7}{2}.$$

This gives

$$\mathbb{E}(X) = \sum_{i=1}^n \mathbb{E}(X_i) = n \left(\frac{7}{2}\right) = \frac{7n}{2}.$$

- (7) An urn initially contains one red and one blue ball. At each stage, a ball is randomly chosen and then replaced along with another of the same colour. Let X denote the selection number of the first chosen ball that is blue. For instance, if the first selection is red and the second blue, then X is equal to 2.

- Find $P\{X > i\}$, $i \geq 1$.
- Show that, with probability 1, a blue ball is eventually chosen. (That is, show that $P\{X < \infty\} = 1$.)
- Find $\mathbb{E}[X]$.

Solution:

- (a) The event $X > i$ means that the first i choices are all red, and

$$P\{X > i\} = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdots \frac{i}{i+1} = \frac{1}{i+1}.$$

This is because a red ball is picked in the first step, and it will be returned into the urn together with an additional red ball, so the urn now contains 2 red balls and 1 blue ball. The probability that a red ball is picked in the second step is thus $2/3$. This procedure continues, and at the beginning of the i -th step there are i red balls and 1 blue ball in the urn.

(b) It follows from

$$P\{X = \infty\} = P\left(\bigcap_i \{X > i\}\right) = \lim_{i \rightarrow \infty} P\{X > i\} = 0$$

that $P\{X < \infty\} = 1$.

(c) Note that $P\{X = 1\} = 1/2$, as there are only two balls at the beginning and the probability of taking a blue ball is $1/2$. When $i > 1$, $P\{X = i\} = \left(\frac{1}{2}\right) \left(\frac{2}{3}\right) \cdots \left(\frac{i-2}{i-1}\right) \left(\frac{1}{i}\right)$ and hence

$$\mathbb{E}[X] = \sum_{i=1}^{\infty} i \cdot P\{X = i\} = \sum_{i=1}^{\infty} i \left[\left(\frac{1}{2} \cdot \frac{2}{3} \cdots \frac{i-2}{i-1}\right) \cdot \frac{1}{i} \right] = \sum_{i=2}^{\infty} \frac{1}{i} = +\infty.$$

(8) A group of 6 female managers and 19 male managers apply for an assignment. A random sample of 5 people is drawn from a hat (without replacement). What is the probability 4 of the chosen managers will be female and 1 will be male?

Solution: It is a hypergeometric distribution, and the wanted probability is

$$\frac{\binom{6}{4} \binom{19}{1}}{\binom{25}{5}} \approx 0.00536.$$

Assignment Questions

- (1) An interviewer is given a list of people she can interview. If the interviewer needs to interview 5 people, and if each person (independently) agrees to be interviewed with probability $\frac{2}{3}$, what is the probability that her list of people will enable her to obtain her necessary number of interviews if the list consists of
- (a) 5 people and
 - (b) 8 people?
- For part (b), what is the probability that the interviewer will speak to exactly
- (c) 6 people and
 - (d) 7 people on the list?

Answer: (a) Let X be the random variable that marks the number of people who agreed to be interviewed. Then $n = 5$. As she needs to interview 5 people, she need to interview all people on the list, and hence the probability that her list of people will enable her to interview 5 people is

$$P\{X = 5\} = \left(\frac{2}{3}\right)^5 = 0.132.$$

(b) Let Y be the random variable that marks the number of people who agreed to be interviewed. For this question, $n = 8$. As she needs to interview 5 people, she need to interview all people on the list, and hence the probability that her list of people will enable her to interview 5 people is

$$P\{Y \geq 5\} = \sum_{k=5}^8 P\{Y = k\} = \sum_{k=5}^8 \binom{8}{k} \left(\frac{2}{3}\right)^k \left(\frac{1}{3}\right)^{8-k} = 0.741.$$

$$(c) P\{Y = 6\} = \binom{8}{6} \left(\frac{2}{3}\right)^6 \left(\frac{1}{3}\right)^2 = 0.273.$$

$$(d) P\{Y = 7\} = \binom{8}{7} \left(\frac{2}{3}\right)^7 \left(\frac{1}{3}\right)^1 = 0.156.$$

- (2) An urn contains N white and M black balls. Balls are randomly selected, one at a time, until a black one is obtained. If we assume that each ball selected is replaced before the next one is drawn, what is the probability that
- (a) exactly n draws are needed?
 - (b) at least k draws are needed?

Answer: Let X denote the number of draws needed to select a black ball, then

$$P\{X = k\} = (1 - p)^{k-1} p$$

where $p = \frac{M}{M+N}$. This gives the answer to part (a):

$$P\{X = n\} = (1 - p)^{n-1} p = \left(\frac{N}{M+N}\right)^{n-1} \left(\frac{M}{M+N}\right) = \frac{M^{n-1} N}{(M+N)^n}.$$

(b)

$$\begin{aligned} P\{X \geq k\} &= \sum_{n=k}^{\infty} ((1 - p)^{n-1} p) = p \sum_{n=k}^{\infty} (1 - p)^{n-1} = p \frac{(1 - p)^{k-1}}{1 - (1 - p)} \\ &= (1 - p)^{k-1} = \left(\frac{N}{M+N}\right)^{k-1}. \end{aligned}$$

- (3) Suppose that independent trials, each having probability p , $0 < p < 1$, of being a success are performed until a total of r successes is accumulated. If we let X equal the number of trials required, then

$$P\{X = n\} = \binom{n-1}{r-1} p^r (1-p)^{n-r},$$

for $n = r, r+1, \dots$. A random variable X whose probability mass function is given as above is said to be a *negative binomial* random variable with parameters (r, p) . Note that a geometric random variable is just a negative binomial with parameter $(1, p)$.

(a) Check that $\sum_{n=r}^{\infty} P\{X = n\} = 1$.

(b) Prove that $\mathbb{E}(X) = r/p$ and $\text{Var}(X) = r(1-p)/p^2$.

(c) If independent trials, each resulting in a success with probability p , are performed, what is the probability of r successes occurring before m failures?

(d) Find the expected value and the variance of the number of times one must throw a die until the outcome 1 has occurred 4 times.

Answer: (a) We let Y be the variable denoting the number of failures before r -th success. Then $P\{X = r + y\} = P\{Y = y\}$ and

$$P\{Y = y\} = \binom{r+y-1}{y} p^r (1-p)^y.$$

Thus,

$$\sum_{x=r}^{\infty} P\{X = x\} = \sum_{y=0}^{\infty} P\{Y = y\} = p^r \sum_{y=0}^{\infty} \binom{r+y-1}{y} (1-p)^y.$$

Note that

$$\binom{r+y-1}{y} = \frac{(r+y-1) \cdots (r+1) \cdot r}{y \cdot (y-1) \cdots 2 \cdot 1} = (-1)^y \frac{(-r)(-r-1) \cdots (-r-y+1)}{y \cdot (y-1) \cdots 2 \cdot 1},$$

and by denoting

$$\frac{(-r)(-r-1) \cdots (-r-y+1)}{y \cdot (y-1) \cdots 2 \cdot 1}$$

by

$$\binom{-r}{y},$$

we have

$$\binom{r+y-1}{y} = (-1)^y \binom{-r}{y}.$$

Thus,

$$\begin{aligned} \sum_{y=0}^{\infty} \binom{r+y-1}{y} (1-p)^y &= \sum_{y=0}^{\infty} \left((-1)^y \binom{-r}{y} \right) (1-p)^y = \sum_{y=0}^{\infty} \binom{-r}{y} (p-1)^y \\ &= (1 + (p-1))^{-r} = p^{-r}. \end{aligned}$$

This gives

$$\sum_{x=r}^{\infty} P\{X = x\} = p^r p^{-r} = 1.$$

(b)

$$\mathbb{E}(Y) = \sum_{k=0}^{\infty} k \binom{r+k-1}{k} p^r (1-p)^k = \sum_{k=1}^{\infty} k \binom{r+k-1}{k} p^r (1-p)^k$$

$$\begin{aligned}
&= \frac{r(1-p)}{p} \sum_{k=1}^{\infty} \binom{r+k-1}{k-1} p^{r+1} (1-p)^{k-1} \\
&= \frac{r(1-p)}{p} \sum_{j=0}^{\infty} \binom{r+1+j-1}{j} p^{r+1} (1-p)^j = \frac{r(1-p)}{p}.
\end{aligned}$$

A similar calculation shows that

$$\text{Var}(Y) = \frac{r(1-p)}{p^2}.$$

As $X = Y + r$, $\mathbb{E}(X) = \mathbb{E}(Y) + r = \frac{r}{p}$, and $\text{Var}(Z) = \text{Var}(Y) = \frac{r(1-p)}{p^2}$.

(c) The solution will be arrived at by noting that r successes will occur before m failures if and only if the r -th success occurs no later than the $(r+m-1)$ -th trial. This follows because if the r -th success occurs before or at the $(r+m-1)$ -th trial, then it must have occurred before the m -th failure, and conversely. Hence, the desired probability is

$$\sum_{n=r}^{\infty} r+m-1 \binom{n-1}{r-1} p^r (1-p)^{n-r}.$$

(d) It is a negative binomial distribution with parameters $r = 4$ and $p = 1/6$. By (b), we have

$$\mathbb{E}(X) = r/p = 24,$$

and

$$\text{Var}(X) = \frac{4 \times (1 - 1/6)}{(1/6)^2} = 120.$$

- (4) Suppose that a batch of 100 items contains 6 that are defective and 94 that are not defective. If X is the number of defective items in a randomly drawn sample of 10 items from the batch, find
- (a) $P\{X = 0\}$ and
 - (b) $P\{X > 2\}$.

Answer: It is a hypergeometric distribution, with $N = 100$, $n = 10$ and $m = 6$, and

$$P\{X = i\} = \frac{\binom{6}{i} \binom{94}{10-i}}{\binom{100}{10}}.$$

Thus, we have

(a)

$$P\{X = 0\} = \frac{\binom{6}{0} \binom{94}{10-0}}{\binom{100}{10}} = \frac{\binom{94}{10}}{\binom{100}{10}} \approx 0.522.$$

(b) $P\{X > 2\} = 1 - P\{X = 0\} - P\{X = 1\} - P\{X = 2\}$

$$= 1 - 0.522 - \frac{\binom{6}{1} \binom{94}{10-1}}{\binom{100}{10}} - \frac{\binom{6}{2} \binom{94}{10-2}}{\binom{100}{10}} \approx 1 - 0.522 - 0.369 - 0.0965 = 0.0125.$$

- (5) A purchaser of electrical components buys them in lots of size 10. It is his policy to inspect 3 components randomly from a lot and to accept the lot only if all 3 are nondefective. If 30 percent of the lots have 4 defective components and 70 percent have only 1, what proportion of lots does the purchaser reject?

Solution: Let X be the event we accept the lot (meaning all 3 components pass inspection), $X_1 = 30\%$ of the lot has 4 defective, and $X_2 = 70\%$ of the lot has 1 defective. Then

$$P(X) = P(X|X_1)P(X_1) + P(X|X_2)P(X_2) = \frac{\binom{4}{0}\binom{6}{3}}{\binom{10}{3}} \times 0.3 + \frac{\binom{1}{0}\binom{9}{3}}{\binom{10}{3}} \times 0.7 = 0.54.$$

Hence the percentage rejected is 0.46.

- (6) Find the expected total number of successes that result from n trials when trial i is a success with probability p_i , $i = 1, \dots, n$.

Answer: Let X denote the number of successes that result from n trials. Also let

$$X_i = \begin{cases} 1 & \text{if trial } i \text{ is a success;} \\ 0 & \text{if trial } i \text{ is a failure.} \end{cases}$$

Then

$$X = X_1 + X_2 + \dots + X_n.$$

This gives

$$\mathbb{E}(X) = \mathbb{E}(X_1) + \dots + \mathbb{E}(X_n) = p_1 + \dots + p_n.$$

- (7) Derive an expression for the variance of the number of successful trials in the question above, and apply it to obtain the variance of a binomial random variable with parameters n and p , and of a hypergeometric random variable equal to the number of white balls chosen when n balls are randomly chosen from an urn containing N balls of which m are white.

Answer: Let X and X_i be the variables in the last question. Then $X = X_1 + X_2 + \dots + X_n$. Then

$$\begin{aligned} \mathbb{E}(X^2) &= \mathbb{E}\left(\sum_{i=1}^n X_i \sum_{j=1}^n X_j\right) = \mathbb{E}\left(\sum_{i=1}^n X_i^2 + \sum_{i=1}^n \sum_{j=1, j \neq i}^n X_i X_j\right) \\ &= \mathbb{E}\left(\sum_{i=1}^n X_i^2\right) + \mathbb{E}\left(\sum_{i=1}^n \sum_{j=1, j \neq i}^n X_i X_j\right) = \sum_{i=1}^n \mathbb{E}(X_i^2) + \sum_{i=1}^n \sum_{j=1, j \neq i}^n \mathbb{E}(X_i X_j). \end{aligned}$$

Now

$$\mathbb{E}(X_i^2) = 1^2 P\{X_i = 1\} + 0^2 P\{X_i = 0\} = 1 \cdot P\{X_i = 1\} + 0 \cdot P\{X_i = 0\} = p_i,$$

and for $i \neq j$, as $X_i X_j = \begin{cases} 1 & \text{if } X_i = 1, X_j = 1; \\ 0 & \text{if one of } X_i \text{ and } X_j \text{ has value 0,} \end{cases}$

$$\mathbb{E}(X_i X_j) = P\{X_i = 1 \text{ \& } X_j = 1\} = P\{X_i = 1\}P\{X_j = 1\} = p_i p_j.$$

(a) If X is a binomial variable, then each p_i is p , we have

$$\mathbb{E}(X^2) = p$$

and

$$\mathbb{E}(X^2) = np + n(n-1)p^2.$$

This gives

$$\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = np + n(n-1)p^2 - n^2p^2 = np - np^2 = np(1-p).$$

(b) If X is hypergeometric, then, each p_i is equal to $\frac{m}{N}$, the probability of draw a white ball (m white balls) from N balls, and given that a white ball is chosen in trial i ,

each of the other $N - 1$ balls, of which $m - 1$ are white, is equally likely to be the j -th ball chosen, for $j \neq i$. Consequently, for $j \neq i$,

$$P\{X_i = 1, X_j = 1\} = P\{X_i = 1\}P\{X_j = 1|X_i = 1\} = \frac{m}{N} \cdot \frac{m-1}{N-1} = p \cdot \frac{m-1}{N-1},$$

and hence

$$\mathbb{E}(X^2) = np + n(n-1)p \cdot \frac{m-1}{N-1}.$$

This gives

$$\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = np + n(n-1)p \cdot \frac{m-1}{N-1} - n^2p^2 = np(1-p) \frac{N-n}{N-1}.$$